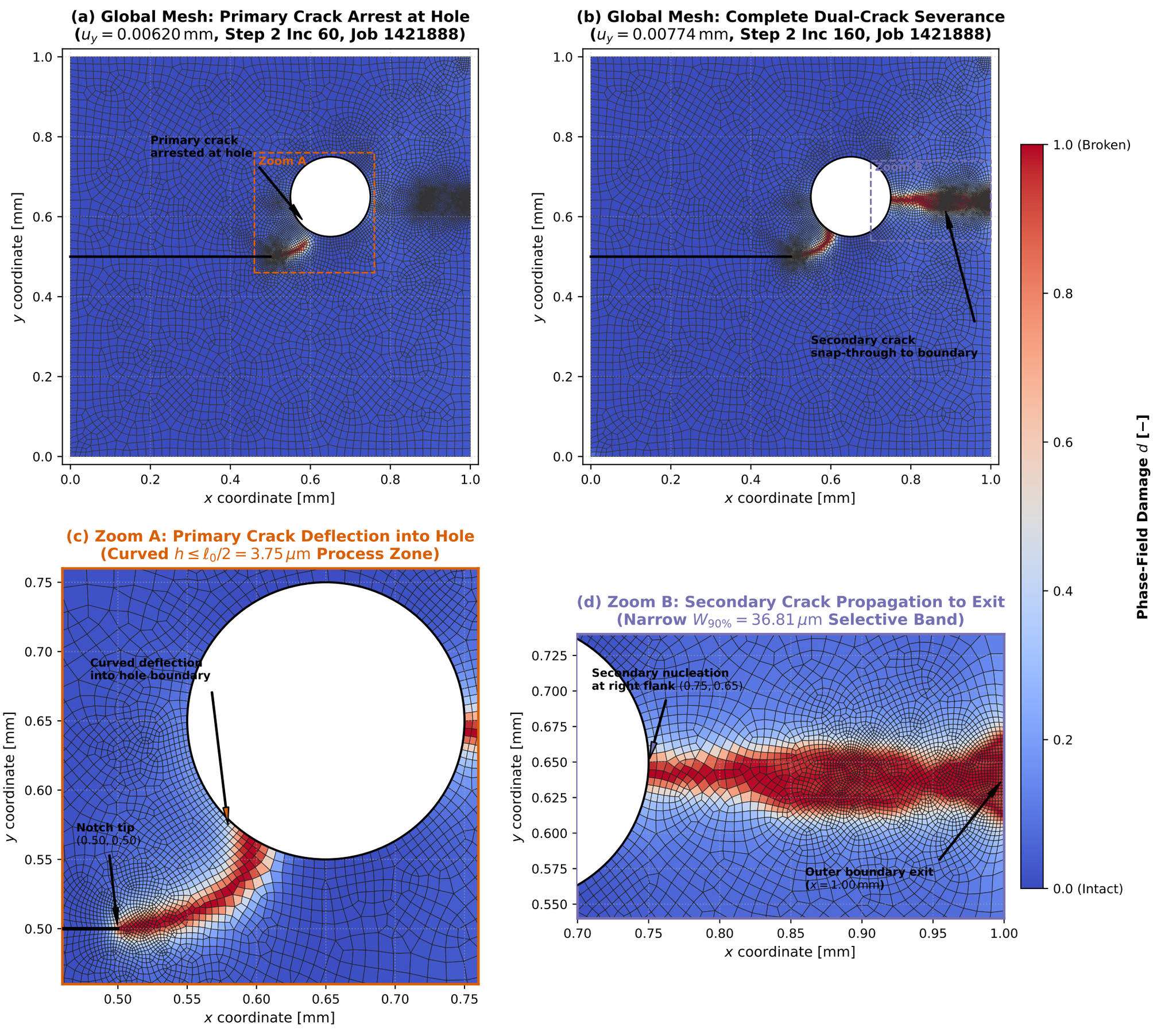

(a) Global Mesh: Primary Crack Arrest at Hole
($u_y = 0.00620$ mm, Step 2 Inc 60, Job 1421888)
Primary crack
arrested at hole
Zoom A
y coordinate [mm]
x coordinate [mm]
(b) Global Mesh: Complete Dual-Crack Severance
($u_y = 0.00774$ mm, Step 2 Inc 160, Job 1421888)
Zoom B
Secondary crack
snap-through to boundary
y coordinate [mm]
x coordinate [mm]
1.0 (Broken)
0.8
0.6
0.4
0.2
0.0 (Intact)
Phase-Field Damage d [–]
(c) Zoom A: Primary Crack Deflection into Hole
(Curved $h \le \ell_0/2 = 3.75\,\mu$m Process Zone)
Curved deflection
into hole boundary
Notch tip
(0.50, 0.50)
y coordinate [mm]
x coordinate [mm]
(d) Zoom B: Secondary Crack Propagation to Exit
(Narrow $W_{90\%} = 36.81\,\mu$m Selective Band)
Secondary nucleation
at right flank (0.75, 0.65)
Outer boundary exit
($x = 1.00$ mm)
y coordinate [mm]
x coordinate [mm]